

Biochemical and Histopathological Features of Porphyria Cutanea Tarda (PCT)

It should be emphasized that patients with clinical signs of PCT may have negative results on a urine fluorescent screening test for porphyrins. In such patients, it is absolutely essential to obtain plasma porphyrin levels and perform quantitative 24-hour urine uroporphyrin (URO) and coproporphyrin (COPRO) determinations, as well as stool protoporphyrin (PROTO) and coproporphyrin (COPRO) measurements. These analyses are preferably performed using high-performance liquid chromatography (HPLC), which often allows differentiation of PCT from variegate porphyria (VP). Rarely, some patients with VP may have normal fecal porphyrin excretion, and bile porphyrin measurements may be helpful in evaluating such cases.

Virtually all patients with PCT have excessive total body iron stores, manifested as increased serum iron, ferritin, and/or hepatocellular iron levels. Occasionally, mild erythrocytosis may be present. An abnormal glucose tolerance test is observed in a minority of patients.

Liver function tests may reveal elevated serum transaminases and γ -glutamyltranspeptidase (GGT) levels, reflecting underlying liver disease. Serum iron and ferritin concentrations are frequently elevated. Measurement of erythrocyte uroporphyrinogen (UROGEN) decarboxylase activity is useful for identifying patients with type II PCT. Mutational analysis is an essential component of the evaluation (see Molecular Genetic Research Strategies for the Porphyrrias).

Histopathology

The characteristic histopathologic finding in PCT is a subepidermal blister (Fig. 132-12). Bullae typically exhibit a corrugated, undulating base described as "festooned." There is little or no inflammatory infiltrate. Periodic acid–Schiff (PAS) staining shows mild thickening of the papillary vessel walls—less

pronounced than in erythropoietic protoporphyria (EPP). Reticulin staining reveals slight proliferation of fibers along the basement membrane.

Direct immunofluorescence studies demonstrate granular deposition of C3 and immunoglobulin G (IgG) at the dermal-epidermal junction and around vessel walls in affected individuals. These findings are most prominent in sun-exposed skin during active disease with high urinary porphyrin excretion and diminish significantly following appropriate treatment. The deposition of immunoglobulins and complement may represent a non-specific response to cutaneous tissue injury. The observed damage to upper dermal vessels and the dermal-epidermal junction supports the hypothesis that porphyrin-mediated photosensitivity leads to the characteristic skin fragility in PCT.

Despite these histopathological features, histologic examination does not substantially contribute to the definitive diagnosis of PCT. This limitation also applies to other cutaneous porphyrias.

Biochemical Features of the Porphyrias

TABLE 132-7

Biochemical Features of the Porphyrias

TYPE OF PORPHYRIA	RBC URO	RBC COPRO	RBC PROTO	PLASMA	STOOL URO	STOOL COPRO	STOOL PROTO	URINE COLOR	URINE ALA	URINE PBG	URINE URO	URINE COPRO
Erythropoietic Congenital Porphyrria	+++ ++	+++++	+++	↑URO & COPRO	+	+++	+	Pink to red	N	N	++++ + ^a	++ ^a
Erythropoietic Protoporphyria	N	N to +	↑↑↑	↑PROTO	N to ↑	N to ↑	↑↑↑	Normal	N	N	N to ↑	N to ↑

Note: ALA = aminolevulinic acid; PBG = porphobilinogen; N = normal; ↑ = increased; + = positive; RBC = red blood cell.